



University of Asia Pacific

Department of Computer Science and Engineering

CSE 450: VLSI Lab

LAB REPORT

Experiment Number: 03

Experiment Title: Design and Simulation of an 8-Bit Register Using D Flip-Flops

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Date of Submission: 27 September , 2026

1. Experiment Name

Design and Simulation of an 8-Bit Register Using D Flip-Flops

2. Objective

- To design and implement an 8-bit register using eight 1-bit D flip-flop based register stages.
- To understand how each bit of input data is stored separately when the clock receives a rising edge.
- To use a common clock signal for all eight register stages and observe how the input values are transferred to the corresponding outputs.
- To synthesize the design in **Xilinx ISE 14.7** and examine the generated RTL schematic.
- To create a testbench and simulate different input patterns in **ISim** to verify that the stored output values match the applied inputs after the clock edge.

3. Apparatus / Hardware & Software Requirements

- **Xilinx ISE 14.7** — used to write the Verilog code, synthesize the 8-bit register, and generate the RTL schematic.
- **ISim Simulator** — used to simulate the register and observe the input, clock, and output waveforms.
- **Verilog HDL** — used to describe the 1-bit register, the complete 8-bit register, and the testbench.
- **Oracle VirtualBox** — used to run the Xilinx ISE virtual machine environment.

4. Circuit Diagram / Schematic

The 8-bit register was created by using **eight separate 1-bit register blocks**. Each block stores one bit of data. The inputs are given separately as **D0 to D7**, and the corresponding outputs are **Q0 to Q7**. All eight register blocks use the same **clock (clk)** signal, so the data is stored according to the clock operation.

The top-level RTL schematic shows the external structure of the 8-bit register, where the eight data inputs, one clock input, and eight outputs are connected to a single `register_8bit` block. The expanded RTL schematic shows the internal structure, where the design is divided into eight individual 1-bit register stages.

Figure 1



Figure 1: Top-Level RTL Schematic of the 8-Bit Register

Figure 1 shows the external connections of the 8-bit register, including the separate inputs D0–D7, the common clk input, and the outputs Q0–Q7.

Figure 2

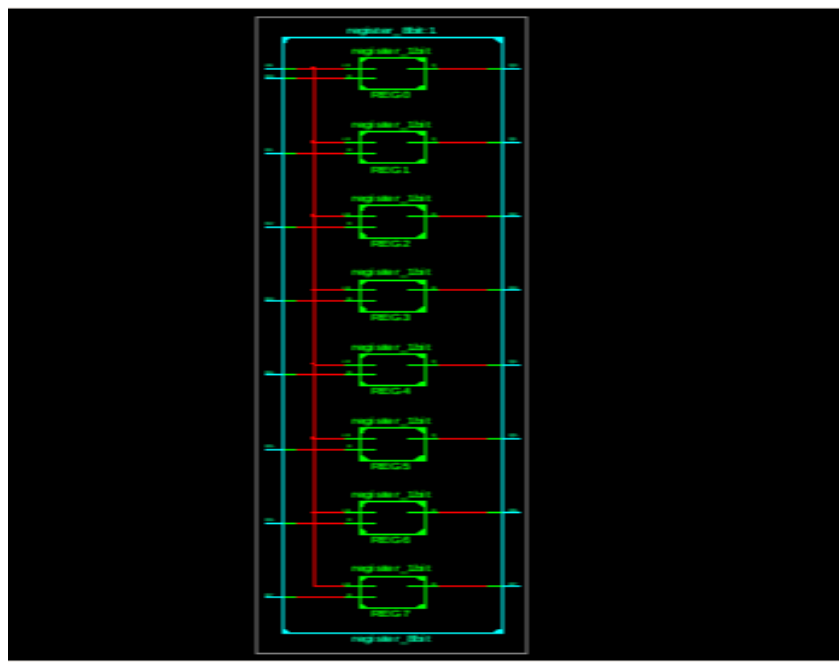


Figure 2: Expanded RTL Schematic of the 8-Bit Register

Figure 2 shows the internal organization of the design, where eight 1-bit register blocks are used together to form the complete 8-bit register.

6. Output Analysis / Observations

The 8-bit register was simulated in ISim using different input patterns. The outputs Q0–Q7 changed only at the **rising edge of the clock**, which shows that the input data was stored correctly by the D flip-flops.

The tested patterns included 00000000, 00000001, 10101010, 11110000, and 11111111. For each case, the stored output matched the applied input after the clock edge.

At the beginning of the simulation, the outputs appeared undefined because the design did not use a reset signal. After the first rising clock edge, the outputs became valid and followed the expected register.



Figure 3: Simulation Waveform of the 8-Bit Register

Table 1: Simulation Observation of the 8-Bit Register

Time (ns)	Input D7–D0	Clock Event	Stored Output Q7–Q0
0–150	00000000	No rising edge yet	Undefined (X)
150–250	00000000	Rising edge at 150 ns	00000000
250–350	00000001	Rising edge at 250 ns	00000001
350–450	10101010	Rising edge at 350 ns	10101010
450–550	11110000	Rising edge at 450 ns	11110000
550–600	11111111	Rising edge at 550 ns	11111111

Before the first rising clock edge, the Q outputs appear undefined (X) because no reset signal is used. At each rising edge, the value present at D7–D0 is stored at Q7–Q0.

7. Result

The 8-bit register was successfully designed and simulated using Verilog HDL in Xilinx ISE 14.7. The simulation showed that the values applied to D0–D7 were stored correctly at Q0–Q7 on the rising edge of the clock. The RTL schematic also confirmed that the design used eight separate 1-bit register stages.

Therefore, the expected operation of the 8-bit register was verified, and the objective of the experiment was successfully achieved.

8. Github Repository Commits

- **Commit:** Add files via upload
- **Commit Link:** <https://github.com/naima-tabassum/CSE450-VLSI-Lab/commit/15a74e011ba9d90917a913fe4610148bf657c46f>

9. Conclusion

This experiment helped me understand how an 8-bit register can be built using eight 1-bit D flip-flop based register stages. I learned how each input bit is stored at its corresponding output when the clock receives a rising edge, while all stages use the same clock signal. I also gained practical experience in creating the Verilog design, generating RTL schematics, writing a testbench, and checking the operation through ISim waveforms. Overall, the experiment improved my understanding of data storage, clock-based operation, and the basic working principle of multi-bit registers.